Project Design Phase-II
Solution Requirements (Functional & Non-functional)

Date	28 October 2026
Team ID	en25cs3010371
Project Name	Electric Vehicle Market Analysis Using Tableau
Maximum Marks	4 Marks

Functional Requirements:

Following are the functional requirements of the proposed solution.

FR No.	Functional Requirement (Epic)	Sub Requirement (Story / Sub-Task)	Priority (High/Medium/Low)
1	Data Access & Integration	The solution should load the EV dataset containing vehicle categories, sales, manufacturers, manufacturing countries, battery types, driving range, charging capacity, efficiency, and environmental metrics.	High
2	Data Processing	The solution should clean and organize the dataset, handle missing values where necessary, and prepare data for analysis in Tableau.	High
3	Business Analysis	The solution should calculate and compare annual sales, average driving range, sales trends, manufacturer performance, battery technology distribution, charging efficiency, and regional environmental impact.	High
4	Business Insights & Analytical Outputs	The solution should provide nine visualizations across three scenarios: Emily's market analysis, David's manufacturer analysis, and Sarah's sustainability analysis.	
5	User Interaction & Analysis	Users should be able to explore charts, view tooltips, compare categories and manufacturers, and analyze trends using Tableau dashboard interactions.	Medium
6	Data Update & Reporting	The solution should allow the dataset to be refreshed when updated data is available and enable users to export or share dashboards and analytical findings.	Medium

Non-functional Requirements:

Following are the non-functional requirements of the proposed solution.

FR No.	Non-Functional Requirement	Description

NFR-1	**Usability**	Dashboards should have clear titles, readable labels, appropriate colors, and an easy-to-understand layout.
NFR-2	**Security**	Project files and dataset access should follow applicable access permissions, and shared links should be configured appropriately.
NFR-3	**Reliability**	Visualizations should display consistent results based on the source dataset and correctly configured calculations.
NFR-4	**Performance**	Dashboards should load and respond efficiently, with optimized worksheets and calculations.
NFR-5	**Availability**	The published Tableau dashboard should be accessible through its sharing link when the hosting service is available.
NFR-6	**Scalability**	The workbook should support additional vehicle records and updated data without requiring the entire dashboard to be redesigned.